

Network Cable and Connectors

1. Draw a labelled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labelling the RJ-45 connector pins.

ANS:

T568B Color Order (RJ-45 Connector)

Pin Number	Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

- Simple RJ-45 Connector Diagram

- RJ-45 Connector (Front View)

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| WO | O | WG | B | WB | G | WBr | Br |

2. List three types of network cables and for each, mention one real-world app or device where it would be used, explaining why.

ANS:

Cable Type	Real-World Device/Application	Reason
Cat5e	WIFI Router	Provides reliable internet connectivity for homes and small offices.
USB Cable	Printer	Connects a printer directly to a computer.
Fiber Optic Cable	IPL Streaming Server	Handles very high bandwidth and long-distance data transmission with minimal signal loss.

3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

ANS:

- Steps to Crimp an RJ-45 Connector
 - Cut the network cable to the required length.
 - Strip about 1 inch of the outer jacket using a cable stripper.
 - Untwist the wire pairs and arrange them according to the T568B color standard.
 - Straighten the wires and trim them evenly.
 - Insert the wires into the RJ-45 connector, ensuring each wire reaches the end.
 - Verify the color order is correct.
 - Place the connector into the crimping tool.
 - Squeeze the crimping tool firmly to secure the connector.
 - Test the cable using a network cable tester.
- Two Common Mistakes to Avoid
 - Using the wrong wire color order (not following T568B).
 - Not pushing the wires fully into the RJ-45 connector before crimping.

- 4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.**

ANS:

Recommended Cable and Connector

- Cable Type: Cat6 Ethernet Cable
- Connector Type: RJ-45 Connector

Explanation

Cat6 cables provide high-speed data transmission up to 1 Gbps or more and reduce interference. RJ-45 connectors are the standard connectors used with Ethernet cables. This combination ensures reliable and stable connectivity for gaming consoles, streaming devices, and display systems during the IPL screening event.

- 5. Find and share a photo (from your home, college, or online) of any network cable or connector in use and describe what type it is and what devices it connects.**

ANS:

Example Description

- I observed a grey Ethernet cable connected to a WiFi router at home.

Cable Type: Cat6 Ethernet Cable

- Connector Type: RJ-45
- Connected Devices: WIFI Router and Desktop Computer

Purpose

- The cable provides a wired internet connection between the router and computer, offering faster speeds and more reliable connectivity than WiFi.

Work

- Sharing a photo from my home, College or online.